Lorie Chen

lorichendd@gmail.com | www.linkedin.com/in/lorie-chen-lovvipop | www.loriechen.com

EDUCATION

Carnegie Mellon University, **School of Computer Science, School of Fine Arts** May 2026

Bachelor of Computer Science and Arts with University Honors

Minors in Media Design and Intelligent Environments

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, 3D Vision, Computational Photography

PROFESSIONAL EXPERIENCE

Research Engineer Intern, Adobe Research May 2025 – August 2025

- Developed a C++ prototype for an AI-assisted video-stabilization feature, optimized memory allocation and linear-algebra operations to reduce processing time by 25%; transitioned prototype into product-development work for future release
- Evaluated integration approaches for 3D Gaussian Splatting in After Effects and authored technical guidance used by a 50+ engineer organization

Research Assistant, Studio for Creative Inquiry, Carnegie Mellon University August 2025 – July 2026

- Integrated 5+ novel computer vision models into custom ComfyUI workflows, reducing manual labeling time, documenting setup and usage for new technical users, supporting generative-AI workshops at CMU and RISD
- Created setup guides and reusable workflows for independent 3DGS capture and processing
- Debugged JavaScript projects and provided technical guidance to 20+ students through office hours, class sessions, and Discord

Research Assistant, Carnegie Mellon University Robotics Institute February 2024 – December 2024

- Developed new robotic animation sequences for CoFRIDA; participants preferred revised behaviors in a user evaluation
- Demonstrated CoFRIDA system improvements to 50+ attendees at IEEE RO-MAN 2024 and communicated technical results during a judged showcase

PROJECT EXPERIENCE

Synth Splat: World Visualizer Tool for DJs and Live Performance July 2026 – Present

- Developing an audio reaction visualization tool for DJs, using generative 3D Gaussian Splats from World Labs' Marble and Meyda.js audio analysis in an agentic workflow for novice coders
- Deployed an interactive prototype at synthsplat.loriechen.com

BXA Capstone: Hidden Point Removal in 3DGS for Heritage Institutions September 2025 – July 2026

- Built a web application for browser-based visualization, segmentation, and annotation of 3D Gaussian Splatting scenes
- Implemented Hidden Point Removal methods to suppress occluded geometry, improving scene readability and making spatial annotation easier for cultural-heritage users
- Deployed an interactive prototype at room.loriechen.com

Cameo Glass Surface Reconstruction, *in collaboration with the Getty Museum* October 2025 – January 2026

- Developed a confocal-stereo 3D reconstruction pipeline for 10+ cameo-glass artifacts, enabling depth maps for evaluation by Getty imaging specialists and cultural-heritage experts
- Translated imaging-research requirements into a repeatable capture and reconstruction workflow for non-Lambertian surfaces

Dead Things in Jars, a virtual museum of wet specimens September 2024 – December 2024

- Built a Python-based 3D Gaussian Splatting capture and rendering workflow for preserving 20+ natural history specimens
- Secured $1,000 in project funding through partnerships with the Center for PostNatural History and Carnegie Museum of Natural History

TECHNICAL SKILLS

Programming: C++, C, Python, JavaScript, GLSL, HTML/CSS

Graphics & Vision: OpenCV, Three.js, spark.js, WebGL, 3D Gaussian Splatting, 3D reconstruction, Computational Photography

AI & Application Development: ComfyUI, Gemini API, React, p5.js, REST/API integration

Developer Tools: Git, Linux, Claude Code, Codex

LEADERSHIP & AWARDS

SIGGRAPH Student Volunteer 2025

Maddy Varner Mastication Grant 2025

Henry Armero Memorial Award for Inclusive Creativity Honorable Mention $250 2026